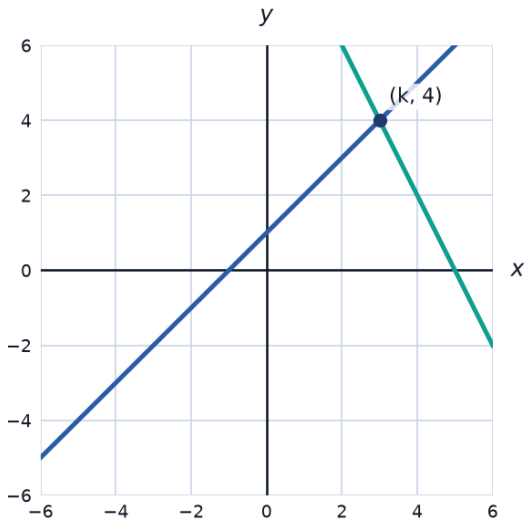
Linear Equations & Lines

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

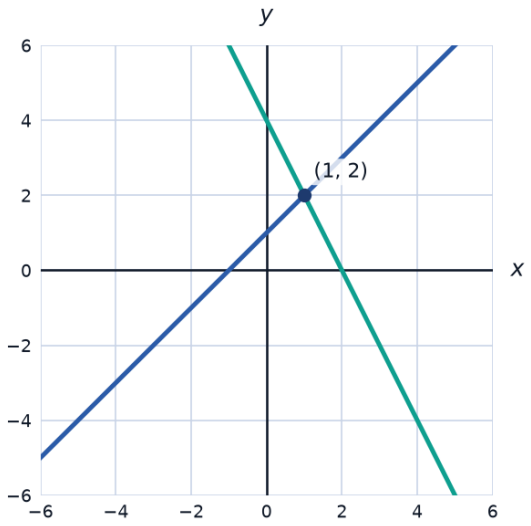
1 [Foundational] • Target: Review: The real number system

Consider the system of linear inequalities represented by the shaded region in the graph below. The boundary lines are given by $y = -2x + 10$ and $y = x + 1$. The point $(k, 4)$ lies on the boundary line $y = -2x + 10$ and is within the feasible region. Determine the value of k.



2 [Foundational] • Target: Review: Integer operations

Consider the system of linear equations graphed below: $\{2x + y = 4; x - y = -1$ What is the solution (x, y) to this system?



3 [Foundational] • Target: Review: Operations with rational numbers

Consider the system of linear equations: $\{2x + y = 7; x - y = 2$ What is the value of x?

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4 [Foundational] • Target: Review: Exponents and radicals

Consider the system of linear equations: $\{2x + y = 5; x - y = 1$ What is the solution (x, y) for this system?

5 [Foundational] • Target: Review: Laws of exponents

Let the rational function be defined as $f(x) = \frac{x^2 - ax + 6}{x^2 - 5x + 6}$. If the graph of $y = f(x)$ has a removable discontinuity (a hole) at $x = 2$, determine the value of the parameter a.

6 [Foundational] • Target: Units resulting from a formula or graph

Consider the function $f(x) = x^2 - 4x + 7$. The graph of $y = f(x)$ is subjected to a sequence of transformations. First, it is translated by a units horizontally and b units vertically. Let this transformed function be $g(x)$. Subsequently, the graph of $y = g(x)$ is reflected in the x-axis to produce the graph of $y = h(x)$. If the vertex of the graph of $y = h(x)$ is at $(-2, -3)$, determine the value of $a^2 + b^2$.

7 [Foundational] • Target: Review: The real number system

The graph of the linear equation $y = mx + c$ is shown below. The line passes through the point $(-3, -11)$ and intersects the y-axis at $(0, k)$. If the slope of the line is -3, determine the value of k.

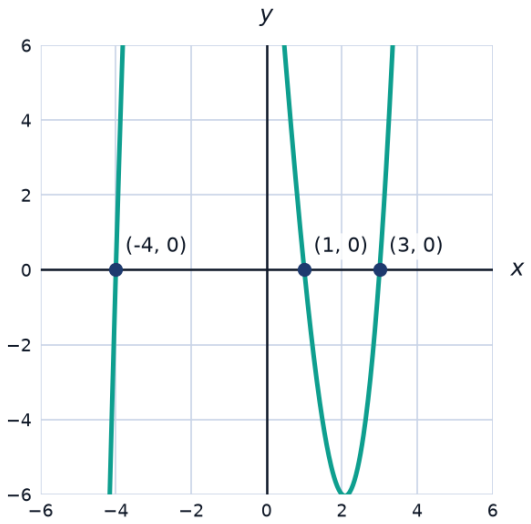
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Review: Integer operations

What is the solution (x, y) to the system of equations below? $x + y, = 8; 2x - y, = 7$

9 [Intermediate] • Target: Review: Operations with rational numbers

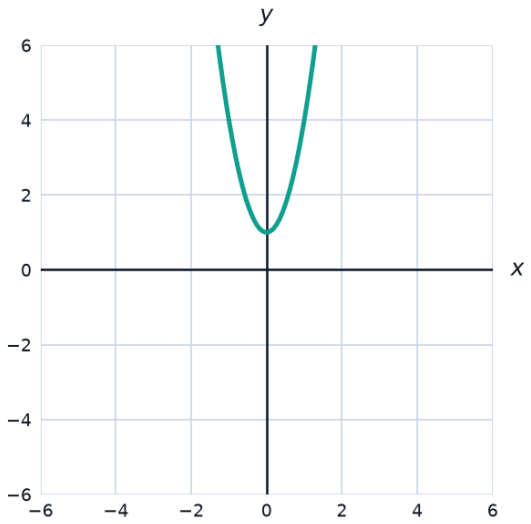
Consider the polynomial $P(x) = x^3 - 13x + 12$. The graph of $y = P(x)$ is shown below. The polynomial can be factored into the form $P(x) = (x - a)(x - b)(x - c)$ where $a < b < c$. Determine the value of $c - a$.



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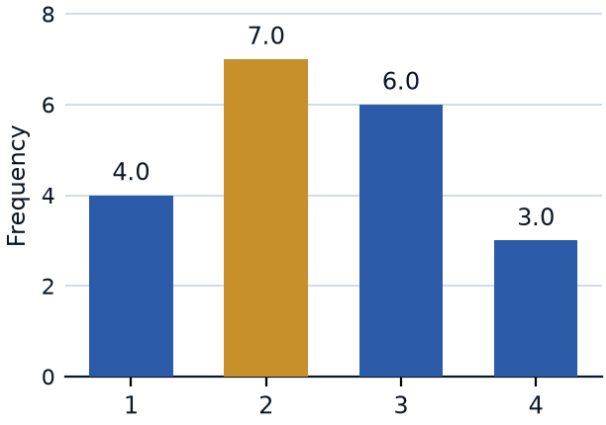
10 **[Intermediate]** • **Target: Review: Exponents and radicals**

The graph of the function $f(x) = ax^2 + b$ is shown below. The graph passes through the points $(-2, 13)$ and $(3, 28)$. Determine the value of b.



11 **[Intermediate]** • **Target: Review: Laws of exponents**

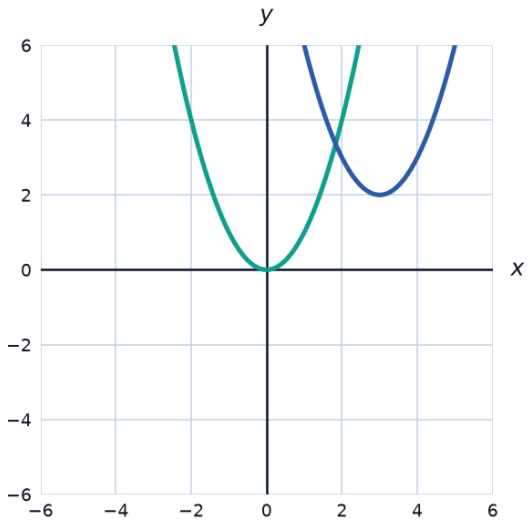
The bar graph shows the frequency distribution of scores obtained by 20 students on a 5-point quiz, where no student originally scored 5. A group of k additional students take the quiz later, and each of these students earns a score of 5. If the mean score of the combined group of $20 + k$ students is equal to 3, what is the value of k?



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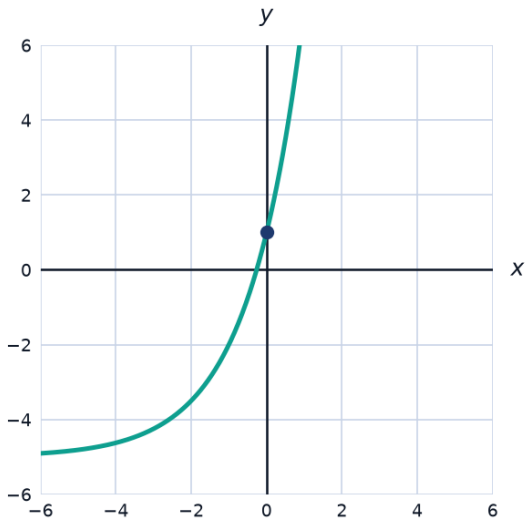
12 **[Intermediate]** • **Target: Units resulting from a formula or graph**

The graph of the function $f(x) = x^2$ is shown below. It is transformed to the graph of $g(x) = (x - 3)^2 + 2$. Which transformation maps $f(x)$ to $g(x)$?



13 **[Intermediate]** • **Target: Review: The real number system**

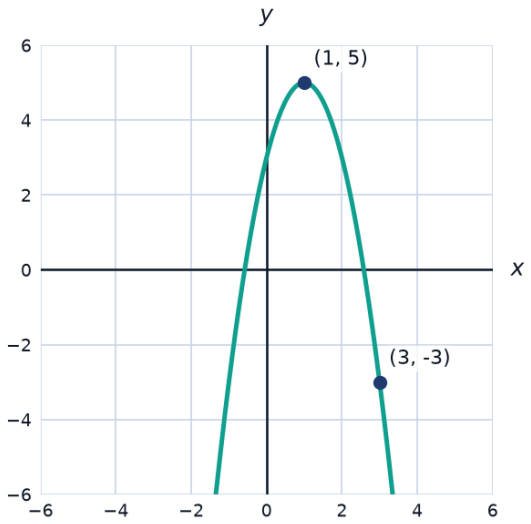
The graph of the function $f(x) = a \cdot b^x + c$ is shown below. The graph passes through the points $(0, 1)$ and $(1, 7)$, and has a horizontal asymptote at $y = -5$. Determine the values of a and b.



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14 **[Intermediate]** • **Target: Review: Integer operations**

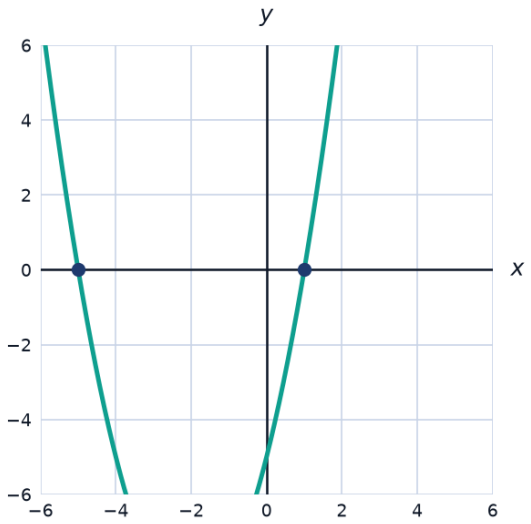
The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(1, 5)$ and the parabola passes through the point $(3, -3)$. Determine the value of a.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Review: Operations with rational numbers** • **Cross-Lesson: 4.01 Functions & Transformations**

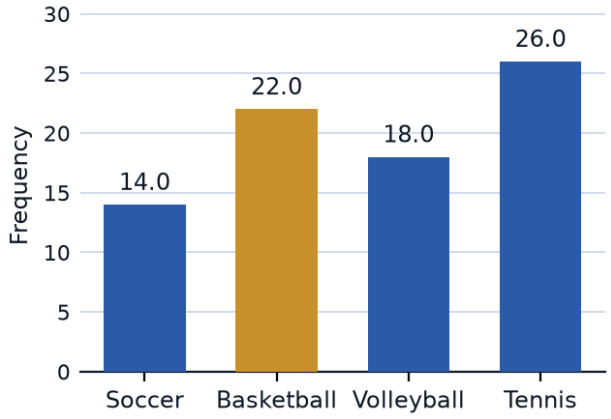
The graph of the quadratic function $f(x) = ax^2 + bx - 5$ is shown below. The function has roots at $x = -5$ and $x = 1$. Find the value of a.



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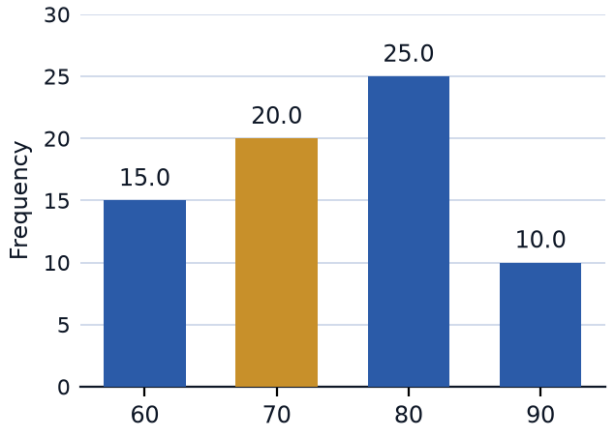
16 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The bar chart below shows the number of students who participated in different sports at a school. What is the total number of students who participated in basketball and volleyball?



17 [Challenging] • Target: Review: Laws of exponents • Cross-Lesson: 6.01 Systems & Inequalities

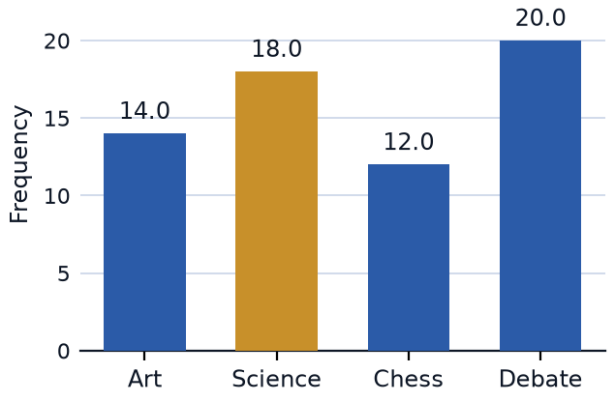
The following bar chart shows the number of students who achieved different scores on a recent quiz. What is the mode of the quiz scores?



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18 [Challenging] • Target: Units resulting from a formula or graph • Cross-Lesson: 7.01 Sequences & Series

The bar chart below shows the number of students in four different clubs at a school. What is the total number of students in the Science Club and the Chess Club combined?



19 [Challenging] • Target: Review: The real number system • Cross-Lesson: 8.01 Factoring Polynomials

Let S be the set of all distinct real solutions to the equation $(x^2 - x + 1)^4 - 10x^2(x^2 - x + 1)^2 + 9x^4 = 0$. What is the sum of the squares of the elements of S?

20 [Challenging] • Target: Review: Integer operations • Cross-Lesson: 9.01 Radicals & Exponents

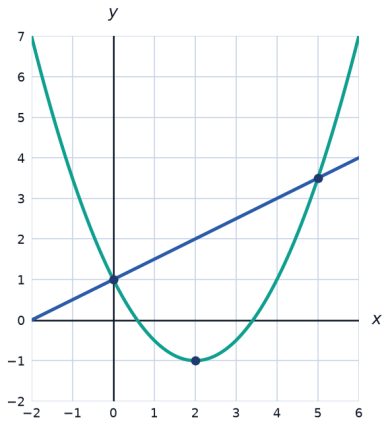
The rational function $R(x) = \frac{x^2 - 7x + k}{x - 3}$ has a removable discontinuity (a hole) at $x = 3$. For what value of k does this occur?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(2)$.



a. 7

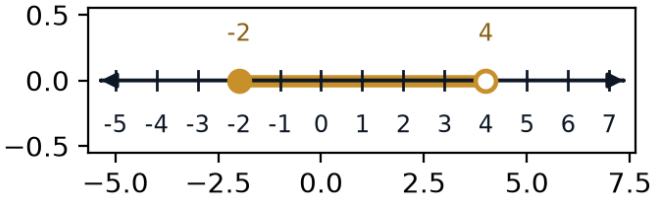
b. 9

c. 4

d. 11

22 [Foundational] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

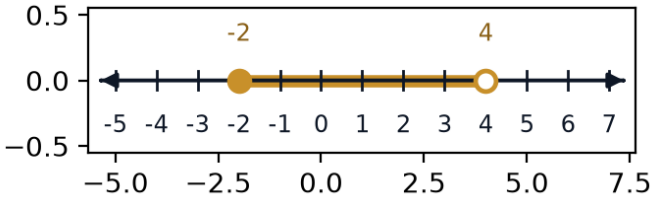
b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

23 [Foundational] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(5)$.



a. 13

b. 15

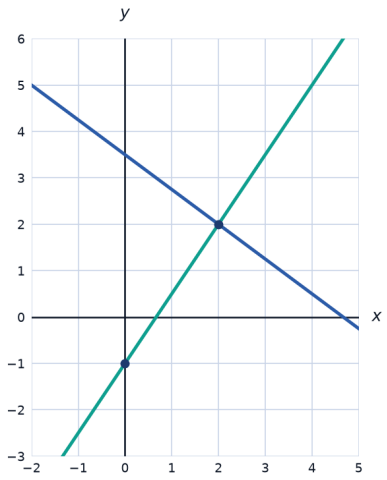
c. 10

d. 17

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24 [Foundational] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(2)$.



a. 10

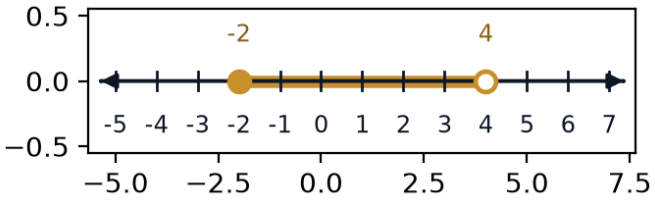
b. 13

c. 6

d. 14

25 [Foundational] • Target: Review: Integer operations

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

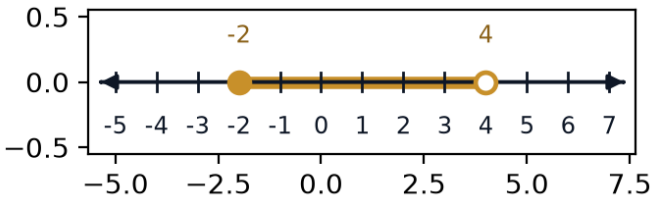
c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

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26 [Foundational] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(5)$.



a. 19

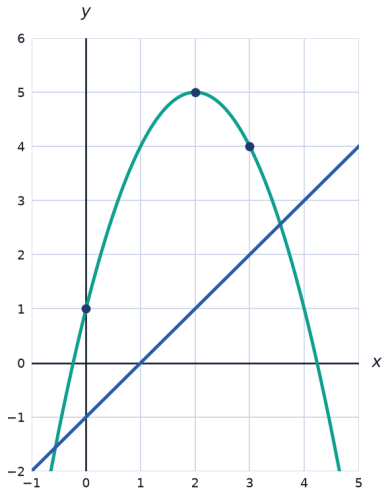
b. 22

c. 15

d. 23

27 [Foundational] • Target: Review: Operations with rational numbers

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(2)$.



a. 13

b. 17

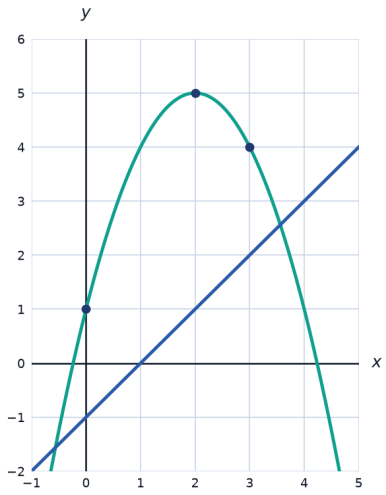
c. 8

d. 17

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28 [Foundational] • Target: Review: Operations with rational numbers

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Review: The real number system

A data set recorded for review: the real number system consists of the four values $\{10, 14, 18, 24\}$.

What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

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31 [Intermediate] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(4)$.

a. 11

b. 13

c. 8

d. 15

32 [Intermediate] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

33 [Intermediate] • Target: Review: Integer operations

A data set recorded for review: integer operations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

34 [Intermediate] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(3)$.

a. 13

b. 16

c. 9

d. 17

35 [Intermediate] • Target: Review: Integer operations

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

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36 [Intermediate] • Target: Review: Operations with rational numbers

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

37 [Intermediate] • Target: Review: Operations with rational numbers

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(3)$.

a. 17

b. 21

c. 12

d. 21

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Review: Operations with rational numbers • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(4)$.

a. 21

b. 25

c. 16

d. 25

39 [Challenging] • Target: Review: Operations with rational numbers • Cross-Lesson: 8.01 Factoring Polynomials

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

40 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 9.01 Radicals & Exponents

A data set recorded for review: exponents and radicals consists of the four values $\{13, 17, 21, 27\}$.

What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

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41 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

42 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 11.01 Statistics & Data Analysis

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

43 [Challenging] • Target: Review: Laws of exponents • Cross-Lesson: 2.01 Quadratic Functions

A data set recorded for review: laws of exponents consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

44 [Challenging] • Target: Review: Laws of exponents • Cross-Lesson: 3.01 Exponents & Polynomials

For the model $f(x) = 6x + 7$ in review: laws of exponents, evaluate $f(4)$.

a. 31

b. 37

c. 24

d. 35

45 [Challenging] • Target: Units resulting from a formula or graph • Cross-Lesson: 4.01 Functions & Transformations

For the model $f(x) = 2x + 8$ in units resulting from a formula or graph, evaluate $f(3)$.

a. 14

b. 16

c. 6

d. 18

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